TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	76.3 ± 2.28
Correlation Stability	88.0 ± 0.89
Distribution Consistency	87.2 ± 0.87
Entropy Preservation	59.7 ± 0.90
Mean Z Score Drift	0.4 ± 0.02
Categorical Fidelity	80.9 ± 2.62
Privacy Risk	39.3 ± 3.20
Data Density	100.0 ± 0.00
Coverage Score	13.2 ± 5.46
Feature Accuracy	61.9 ± 0.83
Novelty Score	0.6 ± 0.02

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	41.724	-3.022	6.8%	OK
TB	3.299	1.106	-2.193	66.5%	Drift
DB	1.486	0.376	-1.110	74.7%	Drift
Alkphos	290.576	221.115	-69.462	23.9%	Warn
Sgpt	80.714	32.292	-48.422	60.0%	Drift
Sgot	109.911	42.089	-67.822	61.7%	Drift
TP	6.483	6.372	-0.111	1.7%	OK
ALB	3.142	3.258	+0.116	3.7%	OK
A/G Ratio	0.947	1.031	+0.084	8.9%	OK
Selector	1.286	2.000	+0.714	55.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	76.3 ± 2.28
Correlation Stability (*2)	88.0 ± 0.89
Distribution Consistency (*3)	87.2 ± 0.87
Entropy Preservation (*4)	59.7 ± 0.90
Mean Z Score Drift (*5)	0.4 ± 0.02
Categorical Fidelity (*6)	80.9 ± 2.62
Privacy Risk (*7)	39.3 ± 3.20
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	13.2 ± 5.46

Feature Accuracy (*10)	61.9 ± 0.83
Novelty Score (*11)	0.6 ± 0.02

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	N/A	0.6743589743589744	-0.010256410256410253	
F1	0.790503612028053	N/A	0.7637598990897891	-0.026743712938263785	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	N/A	0.6914529914529914	-0.011111111111111117	
F1	0.7997903571549948	N/A	0.7825733559094971	-0.017217001245497642	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6606837606837608	N/A	0.6341880341880342	-0.026495726495726502	
F1	0.7642446914350434	N/A	0.7178640384879904	-0.04638065294705307	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	N/A	0.5700854700854701	0.011111111111111127	
F1	0.5587432119380917	N/A	0.5779964057011516	0.019253193763060034	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	N/A	0.5649572649572649	-0.14188034188034188	*
F1	0.8279091624529537	N/A	0.5901224762412812	-0.23778668621167237	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	N/A	0.6376068376068377	-0.03675213675213673	
F1	0.7821343746683586	N/A	0.709113868384741	-0.07302050628361766	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7076923076923076	N/A	0.6683760683760684	-0.03931623931623933	*
F1	0.8104475285030966	N/A	0.7472599425463654	-0.06318758595673118	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	N/A	0.6452991452991453	-0.06068376068376067	*
F1	0.8128002697439015	N/A	0.7052047076416952	-0.10759556210220636	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	N/A	0.635897435897436	-0.04871794871794869	
F1	0.7881619573451331	N/A	0.6947738530610321	-0.09338810428410088	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	N/A	0.6307692307692307	-0.05555555555555556	*
F1	0.7971299319768493	N/A	0.6903120724549551	-0.1068178595218944	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Female	0.8	0.2	143	20	22	6.15796504004058	3.12898252002029	1.02130524399391
								85	45	16
2	39	Female	0.76260713276300	0.2	238	14	16	6.17649985868205	2.67564279657802	0.73756427965780
			42					9	5	25
3	22	Female	0.83412995404225	0.26706497702112	181	15	17	6.30477972425350	3.23890967829575	1.03861838740351
			02	51				1	1	29
4	47	Female	0.7	0.2	164	32	30	7.90171600946342	3.94945394772404	0.99719188598466
								3	6	92
5	8	Male	0.70140862773424	0.16760575182283	296	32	25	5.91549490507674	2.74366327005116	0.80140862773424
			93	283				1	1	93
6	63	Male	1.02981595108101	0.39204542495109	104	18	142	5.00914502023576	3.23598485834963	1.81769481787810
			72	274				6	56	3
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Male	0.7	0.2	162	24	20	6.42262316211668	3.21131158105834	1.0
								2	1	
2	48	Male	1.94142013115643	0.58047337705214	208	47	32	5.76346152458052	3.04881655736963	1.10976331147392
			52	5				9	75	76
3	19	Male	0.62886342931000	0.20962114310333	202	12	13	6.22507486034335	3.10583257413668	0.92886342931000
			53	51				6	6	53
4	45	Male	1.21268438297255	0.37089479432418	208	27	24	6.37836767378558	3.47836767378558	1.18806290497358
			62	543				9	95	4
5	58	Male	0.93623483409056	0.27874494469685	189	19	22	5.43380560833459	3.04878538712201	1.22125505530314
			38	46				9	76	54
6	35	Male	1.94670482371681	0.96676205929204	239	36	146	6.13323794070795	2.83323794070795	0.82664758814159
			94	87				2	15	03
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Male	0.8380097803076616	0.2	194	56	41	7.363784833231445	4.581892416615723	1.6330342310768156
2	5	Male	0.8763285376205988	0.2	337	34	41	7.692270989067784	3.881642688102994	1.0236714623794012
3	60	Male	0.8133689304293305	0.21336893042933036	191	28	35	6.9	3.417825240572440	0.9133689304293304
4	33	Male	1.1714525635350772	0.23572628176753863	490	25	35	6.093894688239342	2.819389468823934	0.8612210623521318
5	35	Male	0.8956913963547751	0.2	190	40	35	7.338777432807024	4.712925810935675	1.7870741890643254
6	84	Female	0.8894301892055820	0.2631433964018607	200	15	17	6.631433964018607	3.452573585607442	1.0368566035981395
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Female	0.7579418097902411	0.1579418097902411	152	21	18	6.768232760839036	3.768232760839056	1.2420581902097589
2	23	Female	1.0933616425413304	0.49004246381200073	190	37	40	7.686723285082674	4.313276714917326	1.2199150723759986
3	14	Male	0.8915683057455117	0.29156830574551174	309	24	18	8.083136611491023	4.166273222982047	0.9915683057455117
4	58	Female	0.6	0.1	187	17	18	6.369110893091076	3.3	1.0516826019739784
5	44	Male	0.7609041993851488	0.2	160	28	25	7.208245407993076	3.451862205533668	0.8781916012297041
6	55	Male	1.3178016622759443	0.4785344415172962	191	51	84	6.821465558482703	3.769633610379323	1.2089008311379732
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Male	0.9078900615291001	0.25394503076455005	193	24	31	6.306604661589949	2.968439753883599	0.84605496923545
2	38	Male	0.994470660075533	0.2972353300377665	216	21	24	7.241941930793097	4.347471270717564	1.4834119802265 9
3	8	Male	0.6	0.1	301	33	40	5.76195549644368	2.641303664295787	0.82024443705546
4	39	Male	0.8370984505032538	0.2370984505032373	115	20	23	6.616649344799711	2.863972348154736	0.7309153754193 78
5	46	Male	0.7331658968803546	0.2	162	30	27	7.568843340555389	3.701506928076808 5	0.9336682062392907
6	45	Male	0.7	0.2	410	25	40	7.2	3.741040975581129	1.0606939837207 53
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	31	Male	0.7113395632403584	0.20850467243026882	189	22	42	7.4085046724302694	3.134018689721075	0.7141744540504 8
2	9	Male	0.8055887094154153	0.2762188937846 4154	296	29	27	5.152437787569283 5	2.029369815630773 6	0.6412603687384 528
3	34	Male	0.9578954351113345	0.2578954351113345	180	44	98	6.926313694665996	3.815790870222669	1.4921111984463 43
4	49	Male	0.9914746379652501	0.2971582126550834	206	23	20	7.234101448139	3.665629830397041 4	1.0291474637965 25
5	27	Male	1.226080678833044	0.4051538041083926	178	54	110	6.710619266507866	3.831650027329544 6	1.8053096332539 331
6	14	Male	0.8345655984882135	0.1781885328294045 3	281	39	42	6.754713320735113	3.563770656588090 4	1.0909426641470 226
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	27	Male	1.0046552974847205	0.2827931784908324	172	58	39	5.748379535472497	2.987448475975553	1.0218621189938883
2	66	Male	0.7	0.2	256	26	28	5.38606264672846	2.0170730878498695	0.5913240145792429
3	45	Male	0.9	0.2	179	27	73	6.601912833523722	3.388695520952327	0.9688695520952326
4	12	Male	0.8	0.2	301	47	66	6.705337230166902	3.501334307541726	1.0991104616388496
5	50	Male	1.0971789653339648	0.3	175	19	19	7.04640034134532	4.454863445343455	1.6858948266698235
6	32	Male	0.9017040771967215	0.2672346923989072	214	18	23	7.3	4.301704077196722	1.4017040771967215
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	54	Male	0.8060128039490465	0.2030064019745324	190	14	17	6.024445583216552	3.093987196050935	1.0545096029617849
2	60	Male	2.1382134902682264	0.9606813119888713	270	45	51	6.150552598871404	2.956169554301612	0.9168508662904837
3	24	Male	1.5460063882690225	0.6230031941345113	179	20	19	7.669009582403533	3.958402555307608	1.0707987223461954
4	33	Female	0.8236889879457889	0.2	202	33	35	8.773598216975799	4.481555060271055	1.0473779758915778
5	66	Male	0.8038121515084484	0.2346040505028161	188	18	24	6.780939242457758	3.319355444469028	0.9307918989943678
6	36	Male	0.6	0.1	274	26	39	5.033983242449329	2.0	0.5440111717004407
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	0.7409132965607355	0.21363776552024516	206	15	29	4.763653186242942	2.168188827601226	0.8
2	40	Male	0.8309795454765576	0.1	488	24	34	5.9	2.5929386364296727	0.8085478977564389
3	35	Male	0.8977387494561291	0.1	487	25	33	5.913567503263225	2.8158287538070906	0.9056531263596772
4	36	Female	1.2326917027462019	0.4	351	149	91	8.04936361227912	4.247438720517725	1.0891027657512662
5	30	Female	0.727956652677018	0.2	193	33	35	7.472043347322982	3.753761589723599	1.0430092717788795
6	48	Female	0.9808047292106482	0.28080472921064814	175	26	54	5.615435327443783	2.8385223929325396	0.9692611964662698
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	63	Male	1.1342883013737426	0.4171441506868714	171	42	45	6.033325787659917	3.1228513219090788	0.9756177109568995
2	17	Male	1.1634079158558495	0.5829210707335851	316	13	27	6.234157858532829	2.392693231660377	0.5975657743886791
3	44	Male	0.7630856259817404	0.2	160	28	25	7.1798868622373275	3.4322293661643037	0.8738287480365112
4	54	Male	0.6	0.1	185	18	23	6.404863533982751	3.095136466017249	0.8951364660172499
5	25	Male	1.8127997990139484	0.7597537533910534	181	21	18	8.059138136868686	4.026215214550264	0.9932922922318422
6	36	Male	0.7796195884567201	0.1477717530740326	156	85	23	6.940760823086558	4.056684740777903	1.3522282469259674
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3